

HAT-P-36b

R-0928

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230405 · created 2026-07-29T10:48:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460039.674 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.028
Transit depth (Rp/Rs)^2	1.4e-05 +/- 0.0021 [%]
Orbital Inclination (inc)	80.6 +/- 2.83
Airmass coefficient 1 (a1)	1.225 +/- 0.032
Airmass coefficient 2 (a2)	-0.18 +/- 0.022
Scatter in the residuals of the lightcurve fit is	2.6 %
Best Comparison Star	None
Optimal Aperture	3.59
Optimal Annulus	8.24
Transit Duration (day)	0.058 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.028 vs 0.126 (deviation 3.44σ)
Duration fit vs published	0.058 d vs 0.0925 d (deviation 1.2σ)
Mid-time O–C	8.6 min
Depth SNR	0.0
Residual scatter	2.6 %

Light curve

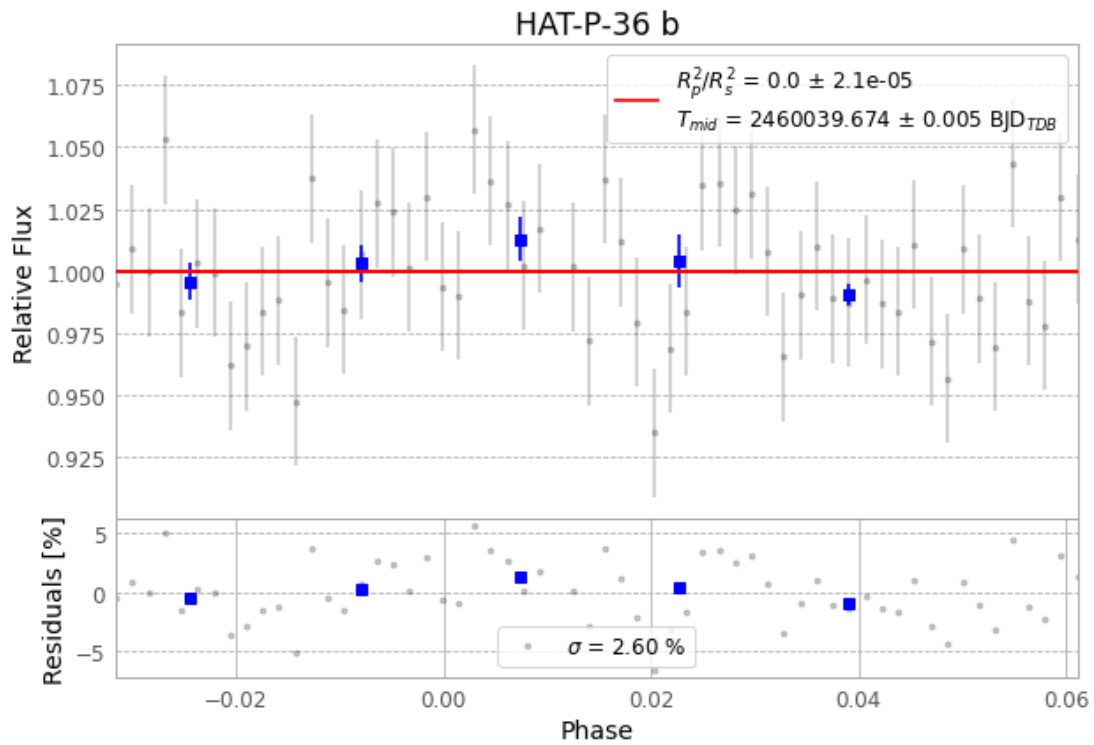


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-04-04.png (SHA-256 87420dc47e1c40e0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [467, 65], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 374b5a6101619702...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

60 FITS frames (39 MB), HATP-36230405030315.FITS → HATP-36230405060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230405.log (8c89ea3e0ce2...), FinalLightCurve_HAT-P-36 b_2023-04-04.png (87420dc47e1c...), FinalLightCurve_HAT-P-36 b_2023-04-04.csv (4d06204d4ffe...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 10:48Z.